Experiment No:3

Metallurgical Sample Preparation

➢ Aim:

Determining the microstructure of the Aluminium and Mild steel

➢ Objectives:

a. Understanding the basic metallurgical sample preparation process
b. Understanding the process of determining the grain size
c. Understand different etching processes

➢ Experimental Set-up:

1. Abrasive cutting machine
2. Hot mounting machine
3. Belt grinder and disc polishing machine
4. Etchant solution
5. Metallurgical microscope

There are four **basic processes** that you will need to become familiar with: sample cutting and sectioning, sample mounting, surface grinding and surface polishing.

- Carefully hold the metal rod on the abrasive cut-off machine and cut a sample of desired dimensions
- Mount the sample using Bakelite powder in a hot-mounting machine set at desired temperature and pressure
- Grind the top surface of the mounted sample using belt grinder to remove the Bakelite and any unevenness
- Wash sample thoroughly after getting a flat sample on the belt grinder
- Polish your sample on the desired abrasive sheet in wet condition (water) for mild steel and dry condition for aluminium
- Rub the sample over the abrasive sheet until all scratches are aligned in a unilateral direction and sample has no uneven region
- Start with abrasive paper of 220 grit, successively increasing to 1000 grit
- Always rotate the sample by 90 degree while going towards the finer abrasive sheet
- After polishing your sample using 1000 grit abrasive sheet, using the double disc polishing machine for further polishing with finer abrasive papers
- Final polishing is performed using the alumina paste of different particle sizes on velvet cloth, successively
- Clean the sample using distilled water and dry in the air
- Prepare suitable etchant as the list provided
- Use a cotton swab and/or a petri dish for the etching. Gently swab the surface of your sample with the etchant for limited time and quantity
- Wash the sample again with distilled water and examine the sample under optical microscope. Several etching cycle might be required to reveal the microstructure
- If the sample is over-etched, repeat the final polishing step and re-etch for a shorter time. Samples to be examined at high magnification generally require shorter etching times than those to be viewed at lower magnifications.

Etching

- Metallographic etching is the process of revealing microstructural details that would otherwise not be evident on the as-polished sample.

- A properly prepared specimen will reveal properties such as grain size, segregation, and the shape, size, and distribution of the phases and inclusions that are present, while other aspects such as mechanical deformation and thermal treatments may also be able to be determined.

Chemical Etching

- This typically involves immersing the sample in an etchant such or swabbing the surface with an etchant.

- The etchant selectively corrodes microstructural features. Immersion time or etching time is highly dependent on the system and in most cases requires experience.

Electrolytic Etching

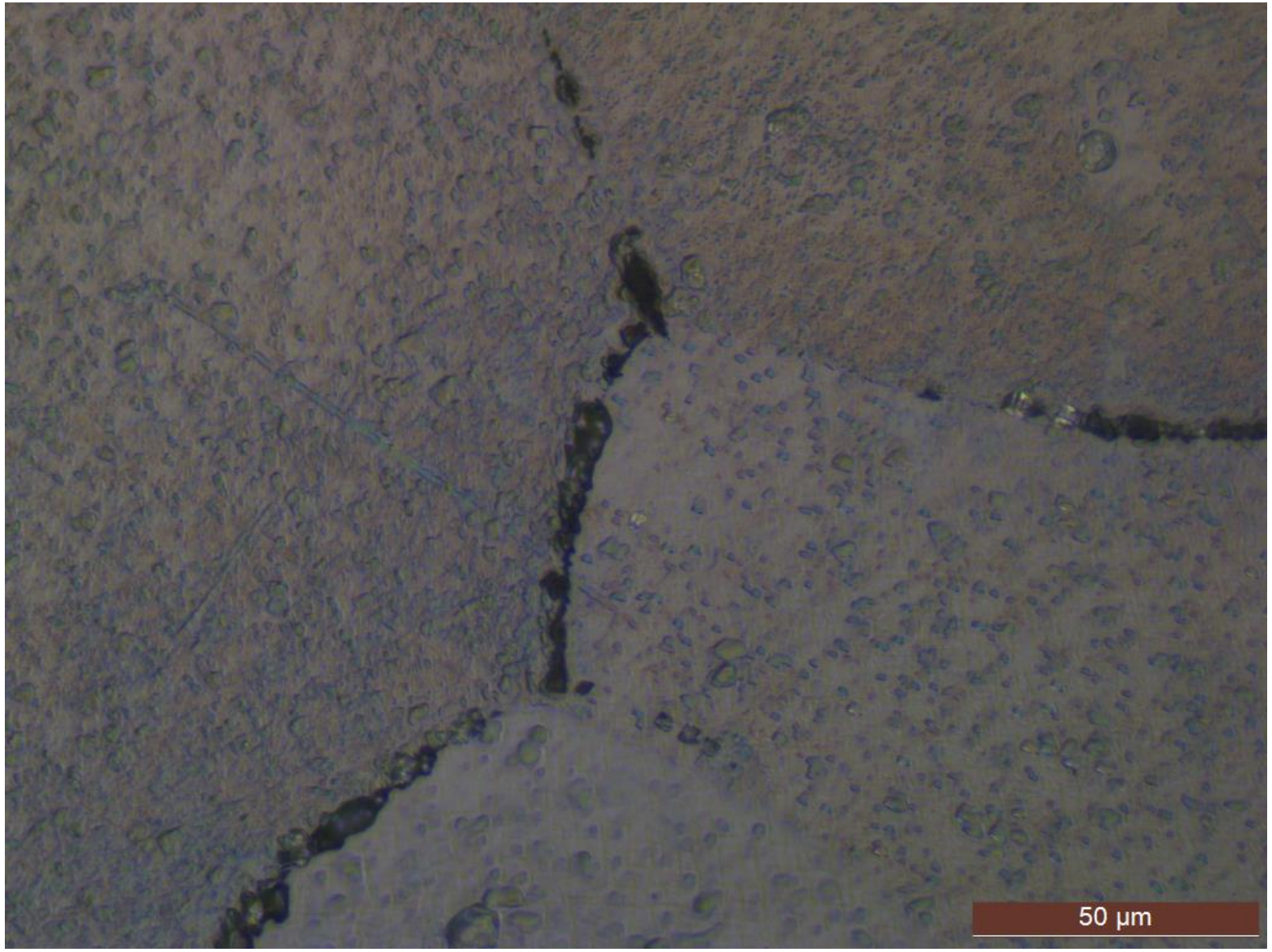
- Electrolytic etching and electropolishing are in effect the same process, except that electrolytic etching uses lower voltages and current densities.

- Most electrolytic etching processes use direct current electrolysis. The process uses the specimen as the anode, with the cathode being a highly insoluble, but conductive material. Typical examples are platinum, graphite and stainless steels.

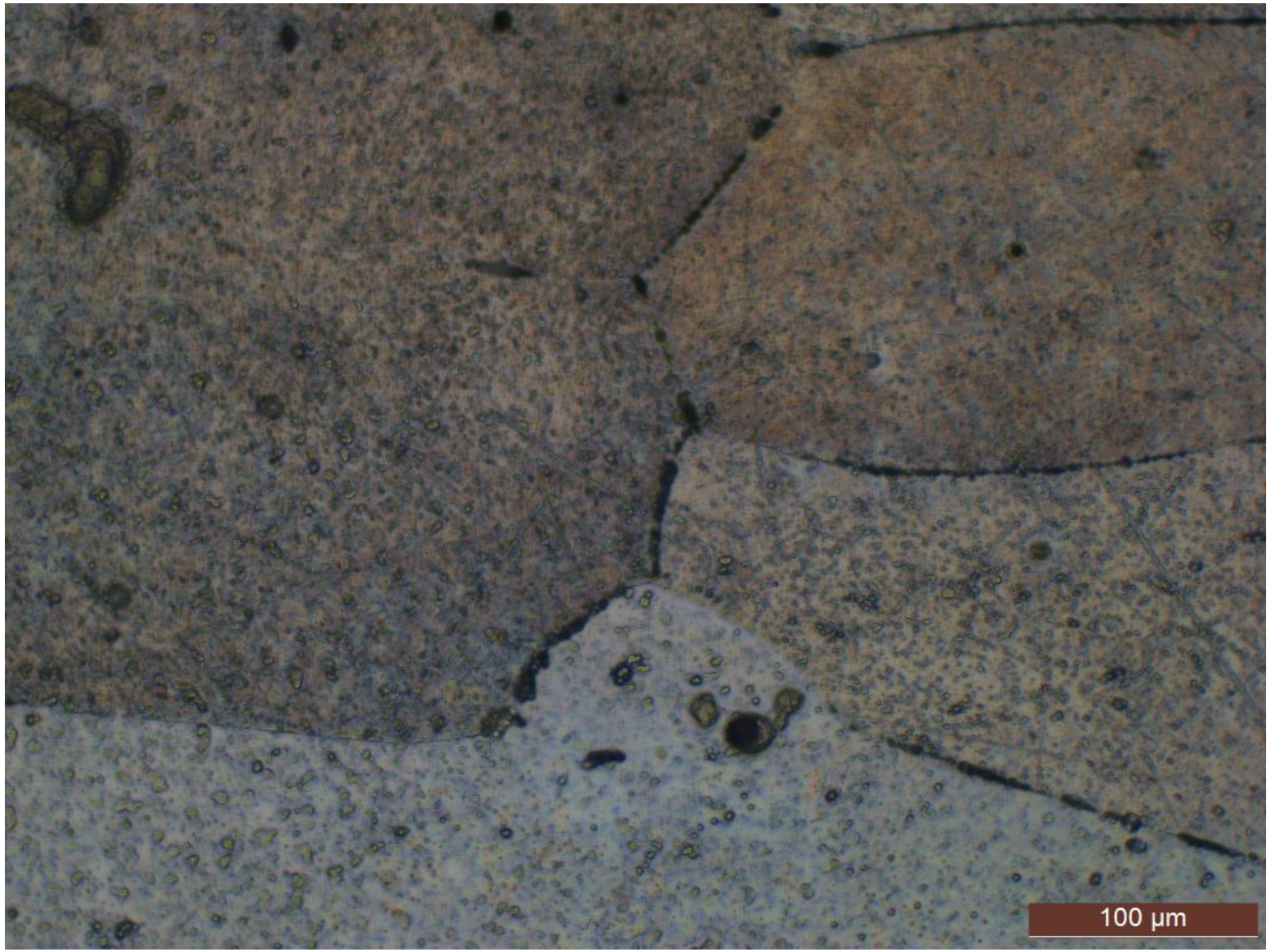
- For **Aluminum**, Etchant prepared as: **5%** Nitic acid (HNO_3)+ **95%** Ethanol

- For **Mild Steel**, Etchant prepared as: **190 ml** of DI water + **5 ml** of Nitic acid (HNO_3)+ **3 ml** of Hydrochloric Acid (HCl) + **2 ml** of Hydrofluoric Acid (HF)

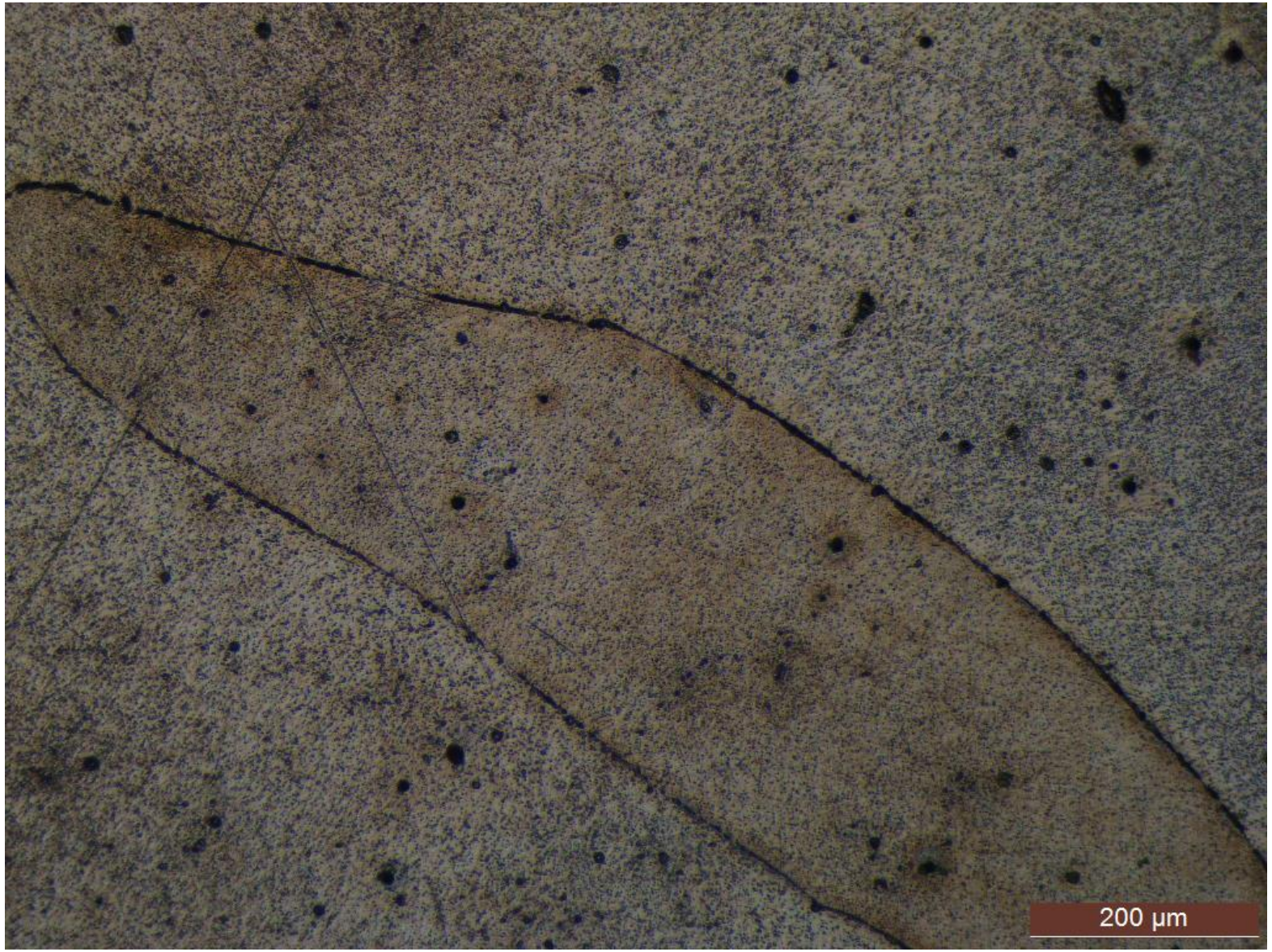
Aluminum Microstructure



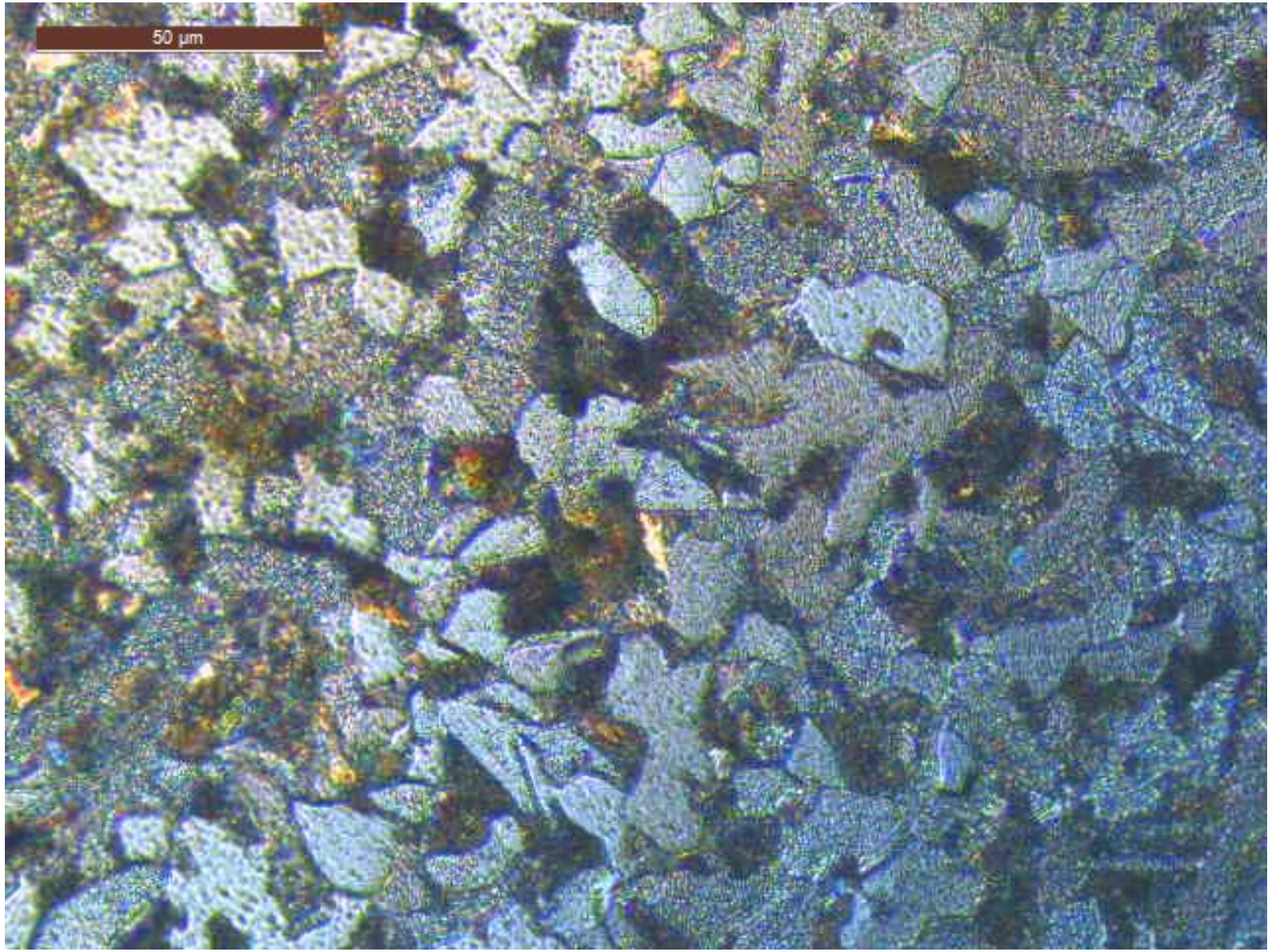
Aluminum Microstructure



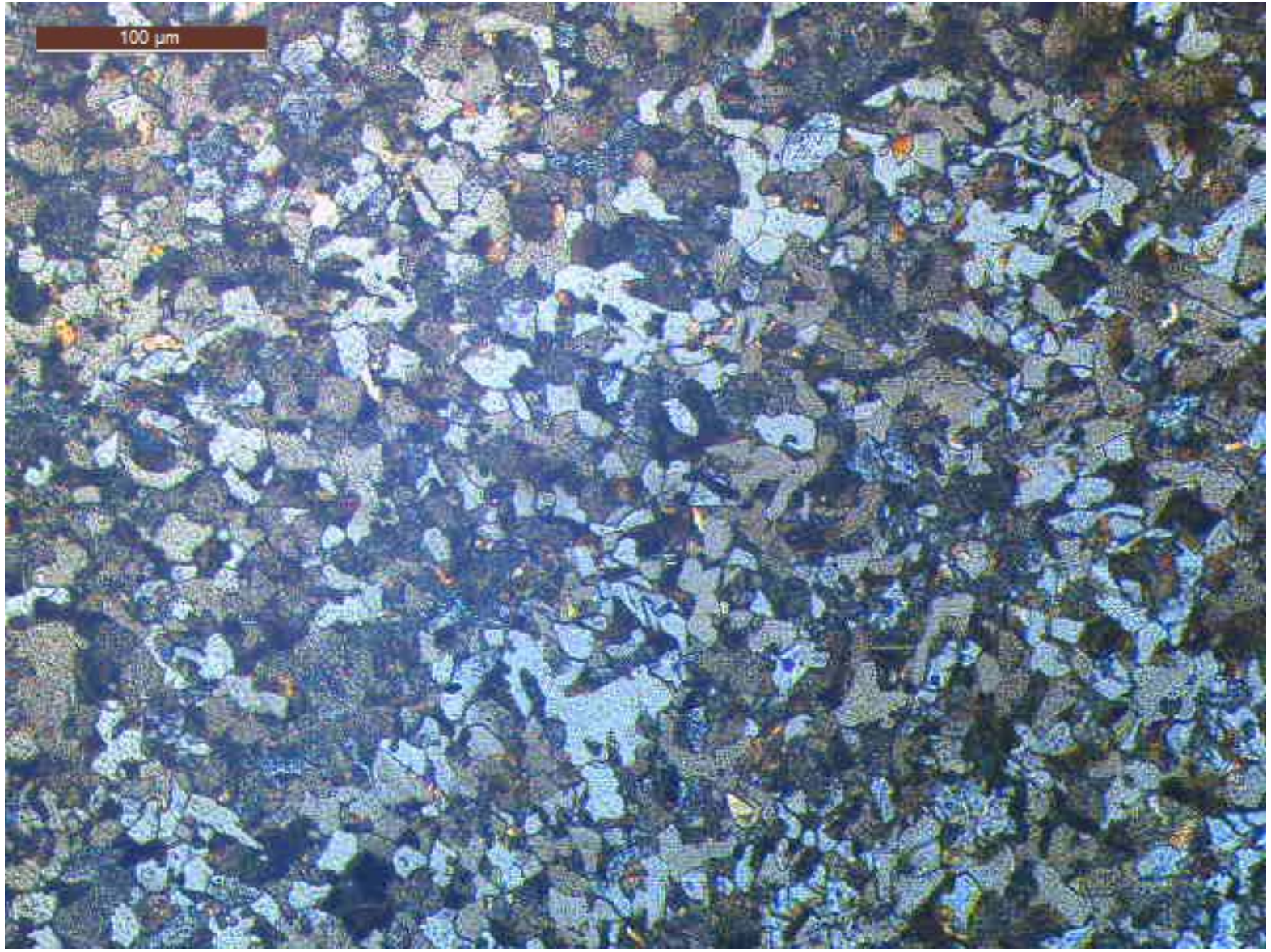
Aluminum Microstructure

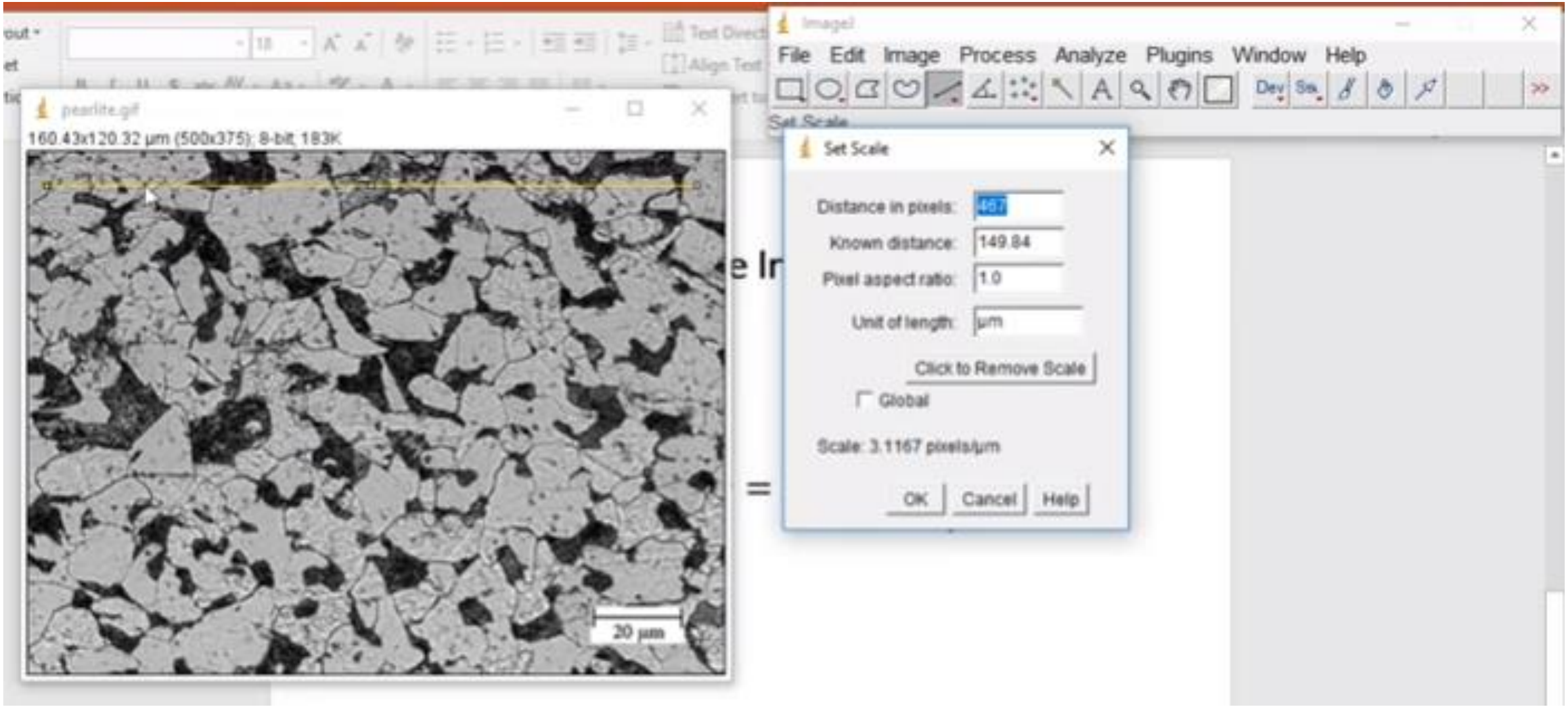


Mild steel Microstructure



Mild steel Microstructure





Average Grain Size (Line Intercept Method)

$$Average\ Grain\ Size = \frac{Line\ Length}{Number\ of\ Grains}$$

→ It is the most common method for grain size measurement.

→ It is devised by American Society for Testing & Materials (ASTM)

$$\boxed{N = 2^{n-1}} \quad \text{—— (I)}$$

N = avg. no. of grains per sq. inch at a magnification of 100x.

n = grain size number.

for magnification other than 100x, eq (I) becomes —

$$\boxed{N_M \left(\frac{M}{100}\right)^2 = 2^{n-1}} \quad \text{—— (II)}$$

N_M = no. of grains per square inch at magnification M.

1) For an ASTM grain size number 5 _____

a. Calculate No. of grains/inch^2 at 100X

2) The ASTM grain size of material which shows 64 grains per inch square at a magnification of 200X is _____

a. 5 b. 6 c. 7 d. 9